Ejercicio 1

Normalizar en FNBC el esquema R(A,B,C,D,E) que verifica el conjunto de dependencias:

$$F = \{A \rightarrow BC,\ BC \rightarrow A,\ BCD \rightarrow E,\ E \rightarrow C\}$$

1- ¿CK?

Ind = ∅

Izq. = D

Der. = ∅

IzqDer = A, B, C, E

Eq = ∅

R (A, B, C, D, E), F = {A->BC, BC->A, BCD->E, E->C}

1: $R_{SI} = R - \emptyset = R$

2: $R_{SIE} = R_{SI} - \emptyset = R_{SI} = R$

$F_{SIE} = F$

3: $K_p = \{D\} = D$

$K^+_{p\,F_{SIE}} = D^+_{F_{SIE}} = \{D\} \neq R_{SIE} \Longrightarrow D \not\in K_{SIE}$

4: $K_p' = K_p \times (\{A, B, C, E\} - K^+_{p\,F_{SIE}}) = \{AD, BD, CD, DE\}$

R (A, B, C, D, E), F = {A->BC, BC->A, BCD->E, E->C}

$$AD^+_{F_{SIE}} = \{A, D, B, C, E\} = R_{SIE} \Longrightarrow AD \in CK_{SIE}$$

$$K_n' = \{BD, CD, DE\}$$

$$BD^+_{F_{SIE}} = \{B, D\} \neq R_{SIE} \Longrightarrow BD \notin CK_{SIE}$$

$$K_p' = \{CD, DE\}$$

$$CD^+_{F_{SIE}} = \{C, D\} \neq R_{SIE} \Longrightarrow CD \notin CK_{SIE}$$

$$K_p' = \{DE, BCD\}$$

R (A, B, C, D, E), F = {A->BC, BC->A, BCD->E, E->C}

$$DE^+_{F_{SIE}} = \{D, E, C\} \neq R_{SIE} \Longrightarrow DE \notin CK_{SIE}$$

$$K_p' = \{BCD, BDE\}$$

$$BCD^+_{F_{SIE}} = \{B, C, D, A, E\} = R_{SIE},$$

$$BCD \in CK_{SIE}$$

$$K_p' = \{BDE\}$$

$$BDE^+_{F_{SIE}} = \{B, D, E, C, A\} = R_{SIE} \Longrightarrow BDE \in CK_{SIE}$$

$$K_p' = \emptyset$$

R (A, B, C, D, E), F = {A->BC, BC->A, BCD->E, E->C}

$CK = \{AD, BCD, BDE\}$

$Ck_{SIE} = \{AD, BCD, BDE\}$

5- $Ck' = Ck_{SIE} \times Ind = \{AD, BCD, BDE\}$

6- $\boxed{Ck = \{AD, BCD, BDE\}}$

R (A, B, C, D, E), F = {A->BC, BC->A, BCD->E, E->C}, CK = {AD, BCD, BDE}

¿R en BCNF? No.

$R_1 = \{A, D, E\}$, $F_1 = \emptyset$, $CK_1 = \{AD\}$, R_1 en BCNF

$R_2 = \{A, B, C\}$, $F_2 = \{A \to BC, BC \to A\}$, $CK_2 = \{A, BC\}$, R_2 en BCNF

$$\boxed{\{A, D, E\}, \{A, B, C\}}$$

R (A, B, C, D, E), F = {A->BC, BC->A, BCD->E, E->C}, CK = {AD, BCD, BDE}

$$\{BCD \rightarrow E \in (F_1 \cup F_2)^+ ?$$

$$E \in BCD^+_{F_1 \cup F_2} \Rightarrow BCD^+_{F_1 \cup F_2} =$$

$$\{B, C, D, A\} \Rightarrow E \notin BCD^+_{F_1 \cup F_2}$$

$$E^+_{F_1 \cup F_2} = \{E\} \Rightarrow C \notin E^+_{F_1 \cup F_2}$$

R (A, B, C, D, E), F = {A->BC, BC->A, BCD->E, E->C}, CK = {AD, BCD, BDE}

$A \to BC \quad BC \to A \quad E \to C$

$R_1 = \{B, C, D, E\}$, $F_1 = \{BCD \to E, E \to C\}$, $CK_1 = \{BCD, BDE\}$, No en BCNF

$R_2 = \{A, B, C\}$, $F_2 = \{A \to BC, BC \to A\}$, $CK_2 = \{A, BC\}$, Si en BCNF

$R_{1.1} = \{B, D, E\}$, $F_{1.1} = \emptyset$, $CK_{1.1} = \{BDE\}$, $R_{1.1}$ en BCNF

R (A, B, C, D, E), F = {A->BC, BC->A, BCD->E, E->C}, CK = {AD, BCD, BDE}

$R_{1,2} = \{E, C\}$, $F_{1,2} = \{E \to C\}$, $CK_{1,2} = \{E\}$

$R_{1,2}$ ea BCNF

$$\boxed{\{A, B, C\}, \{B, D, E\}, \{E, C\}}$$

$$BCD^+_{F_2 \cup F_{1,1} \cup F_{1,2}} = \{B, C, D, A\} \not\supset E$$